

Task 1: Kernel Dimensions

Dominant Kernel: 3×3 Conv2D Dot Product

Dimensions: 3×3 kernel, 1 input channel

Data Types: INT8 weights, INT8 activations, INT32 accumulator

Workload: One invocation computes one output pixel (9 MACs)

Task 2: Total FLOPs

Per Invocation: 9 MACs × 2 = 18 FLOPs per output pixel

Representative Input: 224×224 feature map

Total FLOPs: $224 \times 224 \times 18 = 903,168$ FLOPs

Task 3: Arithmetic Intensity (AI) Bounds

Lower Bound — No Data Reuse

Each invocation (one output pixel) loads all data fresh from off-chip memory:

Traffic = 9 INT8 weights + 9 INT8 activations + 1 INT32 output
= $9 \times 1 + 9 \times 1 + 1 \times 4 = 22$ bytes

AI_{low} = 18 FLOPs / 22 bytes ≈ 0.82 FLOP/byte

Upper Bound — Perfect Weight Reuse (Weight-Stationary Pattern)

Weights are loaded once and reused across all $224 \times 224 = 50,176$ output pixels. Activations and outputs are still streamed from/to off-chip memory.

Weight traffic = $9 \times 1 = 9$ bytes (loaded once)

Activation traffic = $224 \times 224 \times 9 \times 1 = 451,584$ bytes

Output traffic = $224 \times 224 \times 1 \times 4 = 200,704$ bytes

Total traffic = $9 + 451,584 + 200,704 = 652,297$ bytes

AI_{high} = 903,168 FLOPs / 652,297 bytes ≈ 1.38 FLOP/byte

Reuse pattern: Weight-stationary. The 3×3 kernel is held on-chip while all input patches stream through, eliminating repeated weight fetches across output spatial positions.

Task 4: Roofline Sketch

Platform parameters used:

Platform	Peak Compute	Memory BW	Ridge Point	Notes
i5-1135G7 (SW)	268.8 GFLOP/s	51.2 GB/s	5.25 FLOP/byte	SW baseline
sky130A ASIC	0.2 GFLOP/s*	3.2 GB/s**	0.063 FLOP/byte	Accelerator

* Peak compute = 100 MHz × 1 MAC/cycle × 2 FLOP/MAC = 0.2 GFLOP/s (single core)

** Interface BW = AXI4-Stream 128-bit @ 200 MHz = 3.2 GB/s

Both AI bounds (0.82 and 1.38 FLOP/byte) fall to the right of the sky130A ridge point (0.063), placing the accelerator firmly in the compute-bound region of the sky130A roofline. The

projected operating point is at $AI = 0.82$, performance = 0.2 GFLOP/s (compute-limited). The software baseline (measured at $AI = 1.057$, 16.73 GFLOP/s on the i5) sits in the memory-bound region of the i5 roofline.

Task 5: Bottleneck Analysis

Current bottleneck: The accelerator is compute-limited on the sky130A roofline. The single MAC core at 100 MHz delivers only 0.2 GFLOP/s, which is far below what the 3.2 GB/s AXI4-Stream interface could feed. The interface bandwidth is not the bottleneck — the compute throughput is.

Single highest-leverage change: Deploy multiple MAC units in parallel (loop unrolling / spatial parallelism). With 9 parallel MAC units — one per kernel element — the design could process all 9 multiplications in a single clock cycle, increasing peak compute to $9 \times 0.2 = 1.8$ GFLOP/s and bringing the ridge point up to $1.8 / 3.2 \approx 0.56$ FLOP/byte, still below the operating AI of 0.82. This pushes the design into the memory-bound region, where the interface bandwidth becomes the actual limiting resource — the correct design target for a streaming accelerator.